

ActInf Textbook Group ~ Cohort 5 ~ Session 9 (Chapter 4, Part 2) ~ 11/14/2

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 9 (Chapter 4, Part 2) ~ 11/14/2023 | 57:11

okay well welcome back everyone it's

11:14 we're in our second discussion on

chapter

so before we go to the chapter and

questions does anyone have any comments

or

thoughts otherwise we'll just try to

rapidly address as many questions as

possible

okay onwards

then

so in equation 419 what is the meaning

of the Tilda ETA a component of the

Tilda Epsilon subv okay so tildas are

trajectories Tilda

ATA error minus mean minus

observation or external

State sometimes I remember that because

the μ and ETA are like inverted of

each

other okay oh okay yeah yeah right but I

but I I I mean there's various notation

sets for internal and external States

but U for mind Eda for environment

exactly exactly but then

e being for for

error yeah but that's

yeah

cool

okay in equation 415 are f and g

associated with VF big F and EF Big G or

is that just a

coincidence that is just a

coincidence s of Rel yeah I I mean

there's there's no problem with adding

more concordances but not the F and the G
have nothing to do with it I think it's
just for
and then picking the next appropriate
letter Jeremy if you if you want to
we're just starting from the bottom just
trying to hit as many of the questions
as possible or feel free to bring up a
thought
gotcha in the Syntax for describing
[Music]
expectation
no I tried to make these renders a
Tech but I ended up just messing up the
entire column so now it looks good I like
it with the code blocks
um so I can talk through this
question so there's
the there's the expectation and then the
subscript and then
the um square bracket with the function
inside now no nowhere in the text could
I find an explanation of the subscript
and it appears to
me that that subscript is
more or less pretty darn important like it
would I'm not clear whether that
subscript contains the waiting
Factor um
or or what its relation is to the
variables that you're summing
over so like in my example I have an X
and a Y in the subscript P of X I
think so because there are times that
where that subscript contains um
multiple internal variables so what I'm
trying to get clear is what we are
actually summing over and what the

waiting factor is whether the waiting
factor is included in the square
brackets or
not
um I believe that the
subscript is um what you're summing over
like as if it were below the sigma like
from you know $n - 1$ to N
10 and then the waiting is described by
the evaluation of the function across
those those categorical or continuous
values right
well well
my my understanding of the expectation
function is that you
have not
just you have the value at a certain in
um value of x and then you have the
probability of that value and so if you
have a dice a six sided dice the
probability is always one over six but
um for our circumstances there's usually
a higher weight
probability right so
I yeah in this block it's like I'm I'm
just bothered that in that box where
they explain expectation they don't
explain the
subscript um and it's it's causing me a
lot of grief I guess okay I'll
just man go for
it oh did you want to add something
manir okay
yeah box 2.2 could have a definition of
subscript
um here's the probability this is the
waiting here's the value of
 x here's the probability of X weighted

by their probability and here's summing
over values and that's the expectation
of
X but you you're right when you have
two variables that it needs to be
treated a little bit
carefully
okay what is meant what is meant by
Niche simple simply environment of the
agent there's also longer answers I'm
sure people can provide anyone else want
to give a
thought environment of the agent
partially constructed or conformed by
the
agent yeah
maybe possible for them to to modify or
construct
it but but not
necessarily but yes across the
interaction it has to
exist
okay if anyone else wants to to ask a
different question just raise your hand
or go for it otherwise we're just trying
to go to as many of the written
questions as
possible this is in response to the
previous session asking how priors
originate in addition to self-
evidencing optimism bias noted in
Chapter 2 I wanted to mention the notion
of genes and
prior genetic inheritance has been
framed in active as part of Prior belief
that concept itself I was unable to
find it but believe it was suggested by
Paul

Badcock

yes um we've

done evolutionary models and others have
too where genes provide

priors any other comments people want to
make on

this

okay what is a generative model how is
generative

model related to a generative
process anyone have a thought on

okay so the textbook uses this

distinction between a generative model
and a generative process the generative

Model includes the agent model of
observations and also how hidden States
result in those observations so

internally

held variables about external States

um and they distinguish that from

generative process which is the process

that gives rise to the observations that

the agent observes so that's kind of the

um way that the textbook describes

2022 however very importantly in the

subsequent months and

years a lot of people started using

generative model to refer to a system

level description for example here in

guest stream 45 from I guess earlier in

year we had Maxwell and Ma speaking

about this and in the the paper on the

inner screen model of Consciousness with

the quantum Fe they use generative model

now to describe the entire

system um so nothing is functionally

different in terms of the particular

partitioning it's just about whether the

generative model is referred to as the agent and then the generative model is about the generative process or the generative Model includes the whole thing and then you don't even need to mention the generative process um at 44 minutes in the transcript at this point right here Ali asks that exact question to the authors so short answer generative model is coming to be used to describe all of the system level descriptions that the model has access to and then if we're interested in just the blanket and internal States we call that the particular States or the thing if we're interested in just the internal and the active States we call those the autonomous States if we're interested in just the sensor and active states of course they're just the blanket States and then if we're interested in all of the states we can just refer to system any comments or thoughts okay I'm stuck with the derivation of equation 4 left and 413 I wrote which part I'm stuck with in the PDF cool that looks awesome um if anybody wants to go through it we could um but I think let's just save for like I'll just write a comment that it would greatly benefit from somebody who wants to look through this derivation um another nice comment would might be uh in the

I believe it's in the math group one of the first links goes through um the P mdp derivation in more detail cool cool to yeah yeah okay we'll leave this one keep on going but again if anybody wants another question just raise your hand and we'll go for it otherwise just trying to give a thumbs up to as many questions as we can in equation 414 how is the necessary and sufficient condition derived why is the gradient why is the gradient of f being zero equivalent to having a policy which follows the soft Max of negative sign summation of G and F yeah this is a good question the the fully updated this is just one possible reading um when your policy posterior does not have a gradient so you're sampling your policy posterior um at at a stationarity this corresponds to the soft Max um transformed negative G minus F like doing expected free energy updating of your policies is not resulting in any changes to the posterior policy distribution as scaled by the soft Max which is the

shaky hands

parameter So Soft Max just like it does

in a neural network it sharpens or it

blurs distributions so when there's a

low temperature on decision making

absolute zero then a policy that's

slightly

favored will be like upweighted a lot

whereas in the high temperature decision

making setting the differences amongst

policies are

erased but I'm not 100% sure on this one

let's let's leave

it okay 82 83 the equation says that we

minimize free energy through action yeah

we minimize free energy through two

different mechanisms change the world

change your mind

action on the world action on the mind

and only part of the free energy that

depends directly on action is the lowest

level of

prediction

error in this

setting active inference is just

predictive coding plus reflex arcs what

is implied here that action only happens

at the reflex arc

level does anyone have did anyone ask

this or have a thought on this

um I have a

thought sure is

it well in the predictive coding

Paradigm um you're passing errors and so

the errors are the only way that you get

mind

updates um so is that just just saying

that

um the the source of error only happens
level

so here in this section it's the it's a
motor a motor reflex so on the on the
bottom level of the hierarchy we have
like the sensory

motor predictive coding is
doing um predictions of of sensory
consequences and then reflex arcs are
basically um

responding just to move to just to
straight without

planning um or deliberation or
counterfactuals the reflex arcs are just
moving to resolve the

difference so simple motor
systems they're um able to be modeled as
predictive coding on the base layer with
proprioception and action plus higher
level reflex

arcs that basically enact default
policies when the set point is deviated
from how do you read interpret the
higher or I key Dynamics visuals 44 and
what does anyone

think um one thing I would like to know
is what the dot versus the arrow is
insinuating that wasn't described in the
L just little dot yeah yeah good
question I mean

one short answer is if you're interested
in message passing

diagrams check out the live stream 55
series because that is the most modern
message passing

representation and how it's being
displayed in the book is just being done
kind of

didactically but the way that the message passing is done in live stream 55 that's like the real way if you want to study it from an engineering perspective briefly um these images show two ways of looking at hierarchical predictive processing architectures so let's just look at this one here so on the left side we have the kind of all at once hierarchy so we see a Tilda we know that Tilda means across multiple time points so what's happening here is we have these four clusters this is at level l here's level $l+1$ here's level $l+2$ so this is a three-layer predictive processing hierarchy with observations coming in from the bottom Through Time t of Tilda and the highest level prior is like a fixed parameter that's how the highest hyperparameter works because if you could learn or update or Draw the highest hyper prior from another distribution then it would be another level above it so the left representation is showing the highest hyp of Prior and then the two kind of thick levels that have this four-fold Motif involving the um calculation of Errors between predicted States and incoming bottomup observations so this is the all at once view on the right that has been unfolded so now we see observation at t and $t+1$ so now there's three time steps from left to right and there still is the hierarchy from top to bottom any other thoughts or comments on

it

might want to link to the live stream or
reference that as

um a better

source

box 4.1 says the blanket of variable X
are those variables

that are the uh are the variables that

dot dot dot do Dot and the parents of

X 's children the parents of X and's

children are not given a variable are

these not represented the M mark on

blanket formulation if they are where

they show

up

nothing is simply given a variable

whatever the whatever the node is on the

base graph that's the variable used once

the the system of interest is defined by

the modeler then blanket States can be

calculated and the Markoff blanket is

everything you need to make two other

sets of nodes conditionally independent

so that includes not just direct parents

and direct children but also

co-parents I think the specific answer

to the question is it shows up because

the function $\mu_{\text{Greek } p}$ is taken of the

function $\text{Greek } K$

uh in in the variational message par so

that's saying get the parents of all the

children of

X um so it doesn't need its own variable

um because those they're both functions

so so getting the parents of the

children is a double function call okay

okay I think I see yeah yeah the parents

of x_p of X or row of X the variables

are caused by X children and the parents
of X's children like you could have
another letter if you wanted to talk
about that
um

yeah short answer anytime you're
actually using it you would just use the
variable as defined

locally what's the negative diag in
equation

10 um I think it's a

Identity or a matrix diag

diagonalization let's see what we wrote
for line

four and not not every equation has this
but of course everyone is encouraged if
they do want to read and help the
equations to write natural language
descriptions using the onology

I'm not sure if this is accurate but
previously we've written

entropy or expected average log prob is
equal to the negative diagonal Vector of
dotproduct of Matrix a and natural log
a a is the ambiguity mapping so I think

this operation helps us Define the
ambigu the the entropy of the
ambiguity so if a was just the identity
Matrix that would be the minimum
ambiguity if because just

St States observations would map cleanly
to Hidden States if a were completely
like an even

Matrix like no sensory income inputs
could um could be

differentiated you'd have the highest
entropy in equation 42 there's an equal
sign with a triangle above it what does

mean

it means defined by

definition in equation 4.8 in the second

line we are breaking up the

probabilities multiplying each term in

expected values summing their components

and so breaking into the composition of

two expected values my question is why

the Q Tilda can be broken down into two

Q tildas with different arguments didn't

we Define it in 4.7 as a product of Q

and A

P why the qila broke into two CU hilders

with different

arguments I'm not sure which Q Tilda is

being broken

up I see that s Tilda conditioned on π_i

and S Tilda conditioned upon

observations in π_i and those are the

only ones that I

see but the broader point

about developing the um derivations from

this Top Line into all the bottom

lines that sounds like good

work but I'm not sure about this

question because it doesn't look like

you have to split up

anything

in 4.10 does one sum over t to get G

the right hand side has a t indicating

that is calculated at particular time

but G is given by the expectation over

all future

times does one sum over t to get

G

I'm not really sure which right hand

side is being described

here because there's so many lines the

second

line Okay g

subp

what what about

it um this may relate to the difference

between Tao and

T So T is usually used to mean um

simulation time Horizon and too to mean

like agent

time so here the POS the the the the G

calculated on policies which could be

multiple time steps long

is the entropy of the Hidden States on

that policy at that time

plus that sequence of no it's sequence

of times because you're doing a matrix

operation I think it's basically the the

definition of g is uh over a sequence of

to time steps so yeah you're condensing

the the dimension of G is can collapse

because it is by definition over

multiple tow time

steps yeah g has the same G is a

functional that updates the policy prior

into the policy

posterior so

I guess that all the time steps need to

come into play in the evaluation

so each of these but each of these are

at a given time

step but then it's I guess across time

steps but this is definitely not the

fullest explanation of where expected

free energy is

calculated equation 2.6 has a lot more

information on

okay I was unable to understand why the

expected free

energy is minimized by selecting observations that cause a large change in

belief does anyone have a thought on uh there's multiple paragraphs about it but essentially look up epistemic epistemic value anywhere in the text yeah that's definitely one aspect of it which is just equation 2.6

Information

Gain policies that result in sequences observations that have a large impact on inferences about hidden States Through Time ex Tilda

are loaded with high Information Gain high epistemic

value so expected free energy is minimized by selecting observations that cause a large change in belief yes that's half the

picture expected free

energy is contributed to by both the epistemic and the pragmatic value

what's the role of a generative model in active inference well we talked about generative models um

earlier whether we're using the kind of broader sense of generative model is describing all of the systems

dependencies or the narrower sense of generative model like just the agent in relation with a generative process But whichever way you take it generative model is what we build in active

inference no generative model just fun speculative imagination but this is the actual object or modeling

artifact can you clarify why marginal likelihood is called marginal yes great classical history of Statistics when you have a piece of graph paper and you have um events on the rows and columns and the probabilities of those joint events in the cells on the graph paper then when you sum across the side onto the margin of the paper it's called a marginal probability okay we have some questions with previous upvotes that we could look at or we could continue on with these ones that have no upvotes at all does anyone have any comments or thoughts or anything uh I have one comment I I'm still processing the um the the continuous derivation um section um there is one quote that has I'm trying to grock still which is I believe the mode of the let me get my text um it's talking about how here we go the motion of the mode should be the mode of the motion um in explaining the simplifications that allow the uh continuous derivation um here if you search the motion of the mode should be the mode of the motion yeah this is all related this is leading up to it okay okay um and I'm wondering if there are any sort of resources to break down um the derivation of this section in

more
detail similar to how there was with
with the p
DP par mdp yes the most relevant
in basic Mechanics for stationary
processes this question of hierarchical
basian
stationeriness arises in the generalized
coordinates of
motion so when you have the generalized
coordinates of motion position you know
and higher
derivatives
and you have a
stationeriness in that architecture when
the rate of change is zero on two
different levels that are touching each
other that is the situation that's being
addressed
here but this is I feel like if I find a
graph um presentation of what's being
discussed I might be able to get it so
yeah I'll check that out 26
26 um
here are the generalized
coordinates and this is the continuous
time I'm I don't remember what that yeah
not sure how helpful but that at least
is in the B yeah that gives me something
thank you
yep anyone else want to ask a question
okay let's just try to do a few
more basing brain helps us frame
problems that an agent engaging in
active inference must solve broadly
these are problems of inferring states
of the world perception or sense making
and inferring a course of action

planning or decision selection other than perception and action planning are there other tasks or challenges at the brain or organ mechanisms engage in how would we

know one approach here is just to say well what are you trying to model just name your

phenomena and then maybe that can be cast as a perception or as a planning task or just more broadly as a sense making or decision selection

task that's possible because of course inbound and outbound statistical dependencies are the fundamental across the blanket so a huge variety of phenomena will be addressed by that but there's no reason to think that that's all that there

is it just depends what kind of agent you're trying to model so it's a little bit too General to talk about all agents

but are there tasks or challenges that the brain or organisms engage in just remains to be seen by somebody who wants to list out those phenomena okay for the graphical models in figure 4.2 what is an intuitive example for each structure this is a great basic

question helping get familiar with the basian graph

formalism like okay I boiled water and it was hot and it made a sound or I took two ingredients and I made one dish but this is kind of the fundamentals for um all the other Bas

graph

discussions okay this is another great
um basic or consolidating question that
people could give thoughts
on what are the definitions of
variational free energy and expected
free energy what roles do they play
individually or together in active
inference anyone want to give a
thought otherwise this is truly a
fundamental active
question makes me think of like a
class variational free energy equation
2.5 it's the real time gauge on how well
things are going variational free energy
is a functional of only prior beliefs Q
variational beliefs and incoming sensory
data so it computes a measure of how
surprising sensory incomes are as a
function of the agents beliefs at that
time so it's like a diagnostic of
real-time sense making expect E_d free
energy transposes that setting and that
variational basian approach into
counterfactuals of Futures that haven't
happened yet so expected free energy
considers sequences of possible
observables that could happen
conditioned upon different actions that
the entity has the agency select amongst
affordances and those policies are
evaluated in terms of the epistemic and
pragmatic
value and then given the policy
posterior the modular can take different
methods for policy selection like you
could just draw from the policy
posterior or you could select the

highest evaluated

policy

so everything can be understood as

having a variational free energy due to

its persistence through time and certain

kinds of actions selection and planning

agents can be understood as doing

something like expected free energy and

live stream 55 generalized free energy

GFE it may swallow them

both so we may look back on

2023 when we needed to differentiate vfe

and

efe it may just be a transient framing

because

GFE has some advantages relative to both

them what is the use of categorical

distributions there statistical

distributions that apply to things that

are categorical

in the

world in message passing is there a

decay of information as the distance

between variable X and the individual

markall blanket constituents increases

is implementation of information Decay

and message passing an option for model

implementation I remember we had some

quite interesting discussions on this in

a

previous

time any thoughts on

basian graphs are topological they're

not geometric so it's not exactly clear

what distance means here but let's just

interpret distance topologically like

how many connections the information has

to transfer through then you could

imagine a network that's
lossless at which point there would
not be a decay of information or
you could imagine a lossy Bayesian graph
where information was
lost. Impaction is of information Decay
implementation yes you can always
develop models of information
Decay however I would caution message
passing is probably not where that would
happen message passing is a very
low level
technique that probably doesn't have any
advantages to Simply have informational
degradation happen
in but ironically or interestingly you
might want to
execute a rigorous message passing
schedule to model information Decay on a
base
graph however you wouldn't want to be
information decay in the message passing
you'd want the message passing to
accurately describe the information
decay of the base
graph to update beliefs about policies
we find the posterior that minimizes the
free energy does posterior at time T become
a prior at time $t + 1$
yes
any thoughts or questions people want to
make few more
here few more down
here
 Q is not a single variable Q is used to
refer to any variational distribution of
choice so it can take on
different characteristics it can be

conditioned upon different things as
well you still may have to marginalize
over different
things or condition upon non- different
things

so is Q

$Tilda$ conditioned upon so is Q of X
conditioned on

P_i or Q of O condition on P_i a separate
variational

quantity which is defined in C 410 as
categorical of

O sub

PT

this one no definitely

not it's a separate variational

quality okay policy selection active

inference automatically balances

exploration

exploitation already this is quite a
contentious claim I

mean does it automatically balance it
such that it's

successful or does it give us a pallet
or an

expressivity to set up systems that can
try to balance it I I prefer the second

one but it's written like the first one

one the balance between epistemic and
pragmatic value depends on basically
among other things the intensity of

C this prior is the waiting factor

between exploration and

exploitation

yes the amplitude of

C relates to how much pragmatic value
isal

calculated for the

EF and thus is one way to tune explore exploit ballots another way to do it is to modify the ambiguity of sensory States what is the difference between some other formulation for cognitive agents that places an arbitrary waiting factor between exploration and the only thing I can think of is e makes possible variational approximation methods for optimization which would be unavailable to an objective function simply awaited some of the terms yeah I mean it would be helpful to provide what cognitive architectures you're um you're thinking of for years we've we've as people have brought up other COG of architectures like I've here and elsewhere I've reviewed them um there are usually a vast number of differences between any given cognitive architectures but as this question highlights one of them would be a traditional way to approach this um problem or tension is to have an explicit waiting factor between exploration exploitation whereas in the in the ACM generative model you don't have an explicit tradeoff Factor you just have the generative model and then in different environmental settings it is going to have different arising behaviors and it can do the variational

approximation methods
but you could have a cognitive
architecture that was like a variational
autoencoder which would enable you to
use variational
methods so it's not that the only
difference is the variational
methods
any kind of last thoughts or
questions I have a quick question when
we say variational methods doesn't it
really essentially mean that we have a
some kind of uh parameterized model that
we're optimizing through some kind of um
optimization procedure
that's so that would apply to so
effectively like um deep learning um
methods are in in in a sense also
right they they could be and many
R most narrowly it's when the
approximation to exact Bas is
used so not all methods use
approximation so not all deep learning
or statistics are variational
statistics however broad categories of
are so using this variational
autoencoder scheme the
input obviously is the sensory put the
lat and space are the cognitive
States and then the output is the action
selection so all of this which is
otherwise like just an
engineering
sketch all of this is kind of
refined into the first principles basian
mechanics particular partition physics
of cognitive systems path of least
action

all of those
features that's the kind of analytical
setup of active
inference then it it but it neatly
touches to existing architectures like
this but again there's also machine
learning architectures that don't use
methods so it's not simply just being
parametric
this is a this is a short page but also
a relevant one energy based
models which is funny because they even
talk about generative
models so reproducing data distributions
using these statistically physics based
methods you know there was a there was a
paper that um I don't think made uh much
of a splash but it's really interesting
in the context of
um somewhat somewhat in the context of
active inference is that uh effectively
every classifier um no matter how it's
trained can be seen as an energy based
model and that was the title of the
paper um your classifier is secretly an
energy based model and and you should
treat it as such um
so I think there's some kind of really
interesting deep connection there
between um even a model that's not
explicitly train using variational
methods if you look at through this
through the lens of that paper then
obviously the equation the the equations
for energy based models they look very
much like um the formalism um of uh
based on free energy principle uh I'm
sure there's some interesting nugget of

gold there to to to that that could be
dug out um for someone a little bit more
vers in this mathematically but it's
interesting that the stuff is just
hiding there

yep totally agreed even when the
notation and the ontology look very
different there's no reason to think
that that's the end of the road yeah so
thank you all see you all next time
thank you so much peace by